

Diag Solutions

1. 1. If $\sin(A - B) = \frac{1}{2}$ and $\cos(A + B) = \frac{1}{2}$, where $0^\circ < A + B \leq 90^\circ$ and $A > B$, find A and B .

Answer:

$$A = 45^\circ, B = 15^\circ$$

Solution:

1. Identify the angles for which sine is $1/2$ and cosine is $1/2$.
2. Solve the resulting linear equations for A and B .

Derivation:

$$\begin{aligned}\sin(A - B) &= \sin 30^\circ \\ \implies A - B &= 30^\circ. \cos(A + B) = \cos 60^\circ \\ \implies A + B &= 60^\circ. \text{Adding the equations, } 2A = 90^\circ \\ \implies A &= 45^\circ. \text{Then } 45^\circ + B = 60^\circ \\ \implies B &= 15^\circ.\end{aligned}$$

Introduction to Trigonometry — Trigonometric Ratios of Specific Angles

2. 2. Show that $\tan A \cdot \tan(90^\circ - A) = 1$.

Answer:

$$1$$

Solution:

1. Apply the complementary angle identity for $\tan$.
2. Simplify the product.

Derivation:

$$\tan(90^\circ - A) = \cot A. \text{LHS} = \tan A \cdot \cot A = \tan A \cdot \frac{1}{\tan A} = 1 = \text{RHS}.$$

Introduction to Trigonometry — Trigonometric Ratios of Complementary Angles

3. 3. If $3 \cot \theta = 2$, find the value of $\frac{4 \sin \theta - 3 \cos \theta}{2 \sin \theta + 6 \cos \theta}$.

Answer:

$$1/3$$

Solution:

1. Find $\cot \theta = 2/3$.
2. Divide numerator and denominator by $\sin \theta$ to express in terms of $\cot \theta$ and substitute.

Derivation:

$$\cot \theta = \frac{2}{3}. \text{Expression} = \frac{\frac{4 \sin \theta}{\sin \theta} - \frac{3 \cos \theta}{\sin \theta}}{\frac{2 \sin \theta}{\sin \theta} + \frac{6 \cos \theta}{\sin \theta}} = \frac{4 - 3 \cot \theta}{2 + 6 \cot \theta}. \text{Substitute } \cot \theta = \frac{2}{3} : \frac{4 - 3(2/3)}{2 + 6(2/3)} = \frac{4 - 2}{2 + 4} = \frac{2}{6} = \frac{1}{3}.$$

Introduction to Trigonometry — Trigonometric Ratios

4. 4. If $\sin(A + B) = 1$ and $\cos(A - B) = \frac{\sqrt{3}}{2}$, where $0^\circ < A + B \leq 90^\circ$ and $A > B$, find the values of A and B .

Answer:

$$A = 60^\circ, B = 30^\circ$$

Solution:

1. Use the trigonometric table values for $\sin$ and $\cos$.
2. Set up the system of linear equations based on the given angles.
3. Solve for A and B by adding and subtracting the equations.

Derivation:

$$\begin{aligned} \sin(A + B) &= 1 \\ \implies A + B &= 90^\circ \quad (1) \\ \cos(A - B) &= \frac{\sqrt{3}}{2} \\ \implies A - B &= 30^\circ \quad (2) \\ (1) + (2) : 2A &= 120^\circ \\ \implies A &= 60^\circ \\ (1) - (2) : 2B &= 60^\circ \\ \implies B &= 30^\circ \end{aligned}$$

Introduction to Trigonometry — Trigonometric Ratios of Specific Angles

5. 5. Evaluate the following expression: $\frac{2 \tan^2 45^\circ + \cos^2 30^\circ - \sin^2 60^\circ}{\sin 30^\circ + \cos 60^\circ}$

Answer:

$$2$$

Solution:

1. Substitute the standard values for $\tan 45^\circ$, $\cos 30^\circ$, $\sin 60^\circ$, $\sin 30^\circ$, and $\cos 60^\circ$.
2. Simplify the numerator and the denominator separately.
3. Perform the final division.

Derivation:

$$\text{Numerator: } 2(1)^2 + \left(\frac{\sqrt{3}}{2}\right)^2 - \left(\frac{\sqrt{3}}{2}\right)^2 = 2 + \frac{3}{4} - \frac{3}{4} = 2$$

$$\text{Denominator: } \frac{1}{2} + \frac{1}{2} = 1$$

$$\text{Result: } \frac{2}{1} = 2$$

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6. 6. If $7 \sin^2 \theta + 3 \cos^2 \theta = 4$, show that $\tan \theta = \frac{1}{\sqrt{3}}$.

Answer:

$$\tan \theta = \frac{1}{\sqrt{3}}$$

Solution:

1. Express $\cos^2 \theta$ as $1 - \sin^2 \theta$ using the identity $\sin^2 \theta + \cos^2 \theta = 1$.
2. Solve the resulting linear equation for $\sin^2 \theta$.
3. Calculate $\tan \theta$ using the values of $\sin \theta$ and $\cos \theta$.

Derivation:

$$7 \sin^2 \theta + 3(1 - \sin^2 \theta) = 4$$

$$7 \sin^2 \theta + 3 - 3 \sin^2 \theta = 4$$

$$4 \sin^2 \theta = 1$$

$$\implies \sin^2 \theta = \frac{1}{4}$$

$$\implies \sin \theta = \frac{1}{2}$$

$$\cos^2 \theta = 1 - \frac{1}{4} = \frac{3}{4}$$

$$\implies \cos \theta = \frac{\sqrt{3}}{2}$$

$$\tan \theta = \frac{\sin \theta}{\cos \theta} = \frac{1/2}{\sqrt{3}/2} = \frac{1}{\sqrt{3}}$$

Introduction to Trigonometry — Trigonometric Identities

7. 7. Prove the following trigonometric identity: $\frac{\tan \theta}{1 - \cot \theta} + \frac{\cot \theta}{1 - \tan \theta} = 1 + \sec \theta \csc \theta$.

Answer:

$$1 + \sec \theta \csc \theta$$

Solution:

1. Convert all terms into $\sin \theta$ and $\cos \theta$.
2. Simplify the denominators by taking common denominators.

3. Apply the identity $(a - b)^3 = a^3 - b^3$ or group terms after simplification.
4. Use the identity $\sin^2 \theta + \cos^2 \theta = 1$.
5. Separate the fraction to reach the final RHS.

Derivation:

$$LHS = \frac{\frac{\sin \theta}{\cos \theta}}{1 - \frac{\cos \theta}{\sin \theta}} + \frac{\frac{\cos \theta}{\sin \theta}}{1 - \frac{\sin \theta}{\cos \theta}} = \frac{\frac{\sin^2 \theta}{\cos \theta (\sin \theta - \cos \theta)}}{1 - \frac{\cos \theta}{\sin \theta}} + \frac{\frac{\cos^2 \theta}{\sin \theta (\cos \theta - \sin \theta)}}{1 - \frac{\sin \theta}{\cos \theta}} = \frac{\frac{\sin^3 \theta - \cos^3 \theta}{\sin \theta \cos \theta (\sin \theta - \cos \theta)}}{\frac{(\sin \theta - \cos \theta)(\sin^2 \theta + \cos^2 \theta + \sin \theta \cos \theta)}{\sin \theta \cos \theta (\sin \theta - \cos \theta)}} = \frac{1 + \sin \theta \cos \theta}{\sin \theta \cos \theta} = \frac{1}{\sin \theta \cos \theta} + 1 = \sec \theta \csc \theta + 1 = RHS$$

Introduction to Trigonometry — Trigonometric Identities

8. **8.** If $\sec \theta + \tan \theta = p$, show that $\frac{p^2 - 1}{p^2 + 1} = \sin \theta$.

Answer:

$$\sin \theta$$

Solution:

1. Substitute $p = \sec \theta + \tan \theta$ into the expression $\frac{p^2 - 1}{p^2 + 1}$.
2. Expand the numerator using $(\sec \theta + \tan \theta)^2 - 1$.
3. Apply the identity $\sec^2 \theta - 1 = \tan^2 \theta$ in the numerator.
4. Expand the denominator using $(\sec \theta + \tan \theta)^2 + 1 = \sec^2 \theta + \tan^2 \theta + 2 \sec \theta \tan \theta + 1$.
5. Use $1 + \tan^2 \theta = \sec^2 \theta$ to simplify the denominator.
6. Cancel common terms to arrive at $\sin \theta$.

Derivation:

$$\begin{aligned} p^2 - 1 &= (\sec \theta + \tan \theta)^2 - 1 = \sec^2 \theta + \tan^2 \theta + 2 \sec \theta \tan \theta - 1 = (\sec^2 \theta - 1) + \tan^2 \theta + 2 \sec \theta \tan \theta = 2 \tan^2 \theta + 2 \sec \theta \tan \theta = 2 \tan \theta (\tan \theta + \sec \theta) \\ p^2 + 1 &= (\sec \theta + \tan \theta)^2 + 1 = \sec^2 \theta + \tan^2 \theta + 2 \sec \theta \tan \theta + 1 = \sec^2 \theta + (\tan^2 \theta + 1) + 2 \sec \theta \tan \theta = 2 \sec^2 \theta + 2 \sec \theta \tan \theta = 2 \sec \theta (\sec \theta + \tan \theta) \\ \frac{p^2 - 1}{p^2 + 1} &= \frac{2 \tan \theta (\sec \theta + \tan \theta)}{2 \sec \theta (\sec \theta + \tan \theta)} = \frac{\tan \theta}{\sec \theta} = \frac{\sin \theta}{\cos \theta} \cdot \cos \theta = \sin \theta \end{aligned}$$

Introduction to Trigonometry — Trigonometric Identities

9. **9.** If $\sin \theta = \frac{1}{2}$, then the value of $\cos \theta$ is

Answer: (C)

Solution:

1. Use the identity $\sin^2 \theta + \cos^2 \theta = 1$
2. Substitute $\sin \theta = 1/2$
3. Solve for $\cos \theta$

Derivation:

$$\begin{aligned} \cos^2 \theta &= 1 - (1/2)^2 = 1 - 1/4 = 3/4 \\ \implies \cos \theta &= \frac{\sqrt{3}}{2} \end{aligned}$$

10. 10. The value of $\tan 45^\circ$ is

Answer: (B)

Solution:

1. Recall the trigonometric table for standard angles
2. Identify the value of $\tan 45^\circ$

Derivation:

$$\tan 45^\circ = 1$$

Introduction to Trigonometry — Trigonometric Ratios of Specific Angles

11. 11. If $\cos A = \frac{4}{5}$, then $\tan A$ is

Answer: (A)

Solution:

1. Use $\cos A = \text{base/hypotenuse} = 4/5$
2. Find perpendicular using Pythagoras theorem: $p = \sqrt{5^2 - 4^2} = 3$
3. Calculate $\tan A = p/b = 3/4$

Derivation:

$$\tan A = \frac{\sqrt{1 - \cos^2 A}}{\cos A} = \frac{3/5}{4/5} = \frac{3}{4}$$

Introduction to Trigonometry — Trigonometric Ratios

12. 12. What is the value of $\sin^2 30^\circ + \cos^2 30^\circ$?

Answer: (C)

Solution:

1. Apply the identity $\sin^2 \theta + \cos^2 \theta = 1$ for any angle θ

Derivation:

$$\sin^2 30^\circ + \cos^2 30^\circ = 1$$

Introduction to Trigonometry — Trigonometric Identities

13. 13. If $3 \cot \theta = 2$, then the value of $\tan \theta$ is

Answer: (D)

Solution:

1. Given $3 \cot \theta = 2$, so $\cot \theta = 2/3$
2. Use the reciprocal relation $\tan \theta = 1/\cot \theta$

Derivation:

$$\tan \theta = \frac{1}{2/3} = \frac{3}{2}$$

Introduction to Trigonometry — Trigonometric Ratios

14. 14. The value of $\frac{\sin 18^\circ}{\cos 72^\circ}$ is

Answer: (B)

Solution:

1. Use complementary angle relation $\cos(90^\circ - \theta) = \sin \theta$
2. Set $\theta = 18^\circ$ so $\cos 72^\circ = \sin 18^\circ$

Derivation:

$$\frac{\sin 18^\circ}{\sin 18^\circ} = 1$$

Introduction to Trigonometry — Trigonometric Ratios of Complementary Angles

15. 15. If $\sec \theta = 2$, then θ is

Answer: (C)

Solution:

1. Recall standard values for secant
2. Identify angle where $\cos \theta = 1/2$

Derivation:

$$\begin{aligned}\sec \theta &= 2 \\ \implies \cos \theta &= 1/2 \\ \implies \theta &= 60^\circ\end{aligned}$$

Introduction to Trigonometry — Trigonometric Ratios of Specific Angles

16. 16. The value of $(1 + \tan^2 \theta) \cos^2 \theta$ is

Answer: (A)

Solution:

1. Use identity $1 + \tan^2 \theta = \sec^2 \theta$
2. Multiply $\sec^2 \theta \cdot \cos^2 \theta$

Derivation:

$$\sec^2 \theta \cdot \cos^2 \theta = \frac{1}{\cos^2 \theta} \cdot \cos^2 \theta = 1$$

Introduction to Trigonometry — Trigonometric Identities

17. 17. If $\sin A = \cos A$, then the value of A is

Answer: (B)

Solution:

1. Set $\sin A = \cos A$
2. Divide by $\cos A$ to get $\tan A = 1$
3. Find angle where tangent is 1

Derivation:

$$\begin{aligned}\tan A &= 1 \\ \implies A &= 45^\circ\end{aligned}$$

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18. 18. The value of $\operatorname{cosec}^2\theta - 1$ is equal to

Answer: (D)

Solution:

1. Use the identity $1 + \cot^2\theta = \operatorname{cosec}^2\theta$
2. Rearrange to solve for $\operatorname{cosec}^2\theta - 1$

Derivation:

$$\operatorname{cosec}^2\theta - 1 = \cot^2\theta$$

Introduction to Trigonometry — Trigonometric Identities

19. 19. Assertion (A): For an angle θ , $\sin^2\theta + \cos^2\theta = 1$. Reason (R): This identity holds true for all values of θ where $0^\circ \leq \theta \leq 90^\circ$.

Answer: (A)

Solution:

1. The Pythagorean identity is a fundamental trigonometric identity.
2. It is valid for all values of θ in the given domain.

Derivation:

$$\sin^2\theta + \cos^2\theta = 1 \text{ for all } \theta$$

Introduction to Trigonometry — Trigonometric Identities

20. 20. Assertion (A): The value of $\tan 45^\circ$ is 1. Reason (R): $\tan\theta = \frac{\sin\theta}{\cos\theta}$ and $\sin 45^\circ = \cos 45^\circ = \frac{1}{\sqrt{2}}$.

Answer: (A)

Solution:

1. Recall the trigonometric table values.

2. Use the quotient identity $\tan \theta = \frac{\sin \theta}{\cos \theta}$ to verify.

Derivation:

$$\tan 45^\circ = \frac{\sin 45^\circ}{\cos 45^\circ} = \frac{1/\sqrt{2}}{1/\sqrt{2}} = 1$$

Introduction to Trigonometry — Trigonometric Ratios of Specific Angles

- 21. 21.** Assertion (A): The value of $\sin \theta$ can be 1.5 for some angle θ . Reason (R): The range of $\sin \theta$ is $[-1, 1]$.

Answer: (D)

Solution:

1. The sine function value is bounded between -1 and 1 .
2. Since $1.5 > 1$, the assertion is false.

Derivation:

$$-1 \leq \sin \theta \leq 1$$

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- 22. 22.** Assertion (A): $\sec^2 \theta - \tan^2 \theta = 1$. Reason (R): $1 + \tan^2 \theta = \sec^2 \theta$.

Answer: (A)

Solution:

1. The identity $1 + \tan^2 \theta = \sec^2 \theta$ is a standard NCERT identity.
2. Rearranging this gives $\sec^2 \theta - \tan^2 \theta = 1$.

Derivation:

$$\begin{aligned} 1 + \tan^2 \theta &= \sec^2 \theta \\ \implies \sec^2 \theta - \tan^2 \theta &= 1 \end{aligned}$$

Introduction to Trigonometry — Trigonometric Identities

- 23. 23.** Assertion (A): If $\cos A = \frac{1}{2}$, then $\sin A = \frac{\sqrt{3}}{2}$. Reason (R): $\sin^2 A + \cos^2 A = 1$.

Answer: (A)

Solution:

1. Use the identity $\sin^2 A = 1 - \cos^2 A$.
2. Substitute $\cos A = 1/2$.
3. Calculate $\sin A = \sqrt{1 - 1/4} = \sqrt{3/4} = \sqrt{3}/2$.

Derivation:

$$\begin{aligned}\sin^2 A &= 1 - \left(\frac{1}{2}\right)^2 = 1 - \frac{1}{4} = \frac{3}{4} \\ \implies \sin A &= \frac{\sqrt{3}}{2}\end{aligned}$$

Introduction to Trigonometry — Trigonometric Identities

24. 24. If $\sin A = \cos A$, where $0 < A < 90^\circ$, find the value of A .

Answer:

$$45^\circ$$

Solution:

1. Divide both sides by $\cos A$ to get $\tan A = 1$.
2. Identify the angle where $\tan A = 1$ in the given range.

Derivation:

$$\begin{aligned}\frac{\sin A}{\cos A} &= 1 \\ \implies \tan A &= 1 \\ \implies A &= \tan^{-1}(1) = 45^\circ\end{aligned}$$

Introduction to Trigonometry — Trigonometric Ratios

25. 25. Evaluate $\frac{\tan 65^\circ}{\cot 25^\circ}$.

Answer:

$$1$$

Solution:

1. Use the complementary angle identity $\cot \theta = \tan(90^\circ - \theta)$.
2. Substitute $\cot 25^\circ = \tan(90^\circ - 25^\circ) = \tan 65^\circ$ and simplify.

Derivation:

$$\frac{\tan 65^\circ}{\tan(90^\circ - 25^\circ)} = \frac{\tan 65^\circ}{\tan 65^\circ} = 1$$

Introduction to Trigonometry — Trigonometric Ratios of Complementary Angles

26. 26. Given $15 \cot A = 8$, find the value of $\sin A$.

Answer:

$$\frac{15}{17}$$

Solution:

1. Find $\cot A = \frac{8}{15}$. Using the ratio $\frac{\text{adjacent}}{\text{opposite}}$, the hypotenuse is $\sqrt{8^2 + 15^2}$.
2. Calculate hypotenuse $= \sqrt{64 + 225} = 17$, then $\sin A = \frac{\text{opposite}}{\text{hypotenuse}} = \frac{15}{17}$.

Derivation:

$$\begin{aligned}\cot A &= \frac{8}{15} \\ \implies \text{hyp} &= \sqrt{8^2 + 15^2} = 17 \\ \implies \sin A &= \frac{15}{17}\end{aligned}$$

Introduction to Trigonometry — Trigonometric Ratios

- 27. 27.** Find the value of $2 \tan^2 45^\circ + \cos^2 30^\circ - \sin^2 60^\circ$.

Answer:

$$2$$

Solution:

1. Substitute the standard values: $\tan 45^\circ = 1$, $\cos 30^\circ = \frac{\sqrt{3}}{2}$, $\sin 60^\circ = \frac{\sqrt{3}}{2}$.
2. Simplify the expression $2(1)^2 + (\frac{\sqrt{3}}{2})^2 - (\frac{\sqrt{3}}{2})^2$.

Derivation:

$$2(1)^2 + \frac{3}{4} - \frac{3}{4} = 2 + 0 = 2$$

Introduction to Trigonometry — Trigonometric Ratios of Specific Angles

- 28. 28.** Prove that $\sec^2 \theta - 1 = \tan^2 \theta$.

Answer:

$$\text{Proof based on identity } 1 + \tan^2 \theta = \sec^2 \theta$$

Solution:

1. Start with the fundamental identity $1 + \tan^2 \theta = \sec^2 \theta$.
2. Subtract 1 from both sides to isolate $\tan^2 \theta$.

Derivation:

$$\begin{aligned}1 + \tan^2 \theta &= \sec^2 \theta \\ \implies \tan^2 \theta &= \sec^2 \theta - 1\end{aligned}$$

Introduction to Trigonometry — Trigonometric Identities

- 29. 29.** If $\sqrt{3} \tan \theta = 1$, find the value of $\sin^2 \theta - \cos^2 \theta$.

Answer:

$$-\frac{1}{2}$$

Solution:

1. Find the value of $\tan \theta$ from the given equation.
2. Determine the angle θ or use the ratio to find $\sin \theta$ and $\cos \theta$.
3. Substitute the values into the expression $\sin^2 \theta - \cos^2 \theta$ and simplify.

Derivation:

$$\begin{aligned}\tan \theta &= \frac{1}{\sqrt{3}} \\ \implies \theta &= 30^\circ. \\ \sin 30^\circ &= \frac{1}{2}, \cos 30^\circ = \frac{\sqrt{3}}{2}. \\ \sin^2 30^\circ - \cos^2 30^\circ &= \left(\frac{1}{2}\right)^2 - \left(\frac{\sqrt{3}}{2}\right)^2 = \frac{1}{4} - \frac{3}{4} = -\frac{2}{4} = -\frac{1}{2}.\end{aligned}$$

Introduction to Trigonometry — Trigonometric Ratios

30. 30. Prove that $\frac{\cos A}{1+\sin A} + \frac{1+\sin A}{\cos A} = 2 \sec A$.

Answer:

Proved

Solution:

1. Take the common denominator $(1 + \sin A) \cos A$.
2. Expand the numerator using algebraic identity $(a + b)^2 = a^2 + 2ab + b^2$.
3. Use the identity $\sin^2 A + \cos^2 A = 1$ to simplify the expression and arrive at the result.

Derivation:

$$\begin{aligned}LHS &= \frac{\cos^2 A + (1 + \sin A)^2}{(1 + \sin A) \cos A} = \frac{\cos^2 A + 1 + 2 \sin A + \sin^2 A}{(1 + \sin A) \cos A} \\ &= \frac{(\cos^2 A + \sin^2 A) + 1 + 2 \sin A}{(1 + \sin A) \cos A} = \frac{1 + 1 + 2 \sin A}{(1 + \sin A) \cos A} \\ &= \frac{2 + 2 \sin A}{(1 + \sin A) \cos A} = \frac{2(1 + \sin A)}{(1 + \sin A) \cos A} = \frac{2}{\cos A} = 2 \sec A = RHS.\end{aligned}$$

Introduction to Trigonometry — Trigonometric Identities

31. 31. If $\tan A = \frac{3}{4}$, find all other trigonometric ratios for angle A .

Answer:

$$\sin A = \frac{3}{5}, \cos A = \frac{4}{5}, \csc A = \frac{5}{3}, \sec A = \frac{5}{4}, \cot A = \frac{4}{3}$$

Solution:

1. Consider a right triangle where $\tan A = \frac{\text{Opposite}}{\text{Adjacent}} = \frac{3k}{4k}$.
2. Use the Pythagorean theorem to find the hypotenuse.
3. Calculate the remaining ratios using the sides of the triangle.

Derivation:

Let $P = 3, B = 4$. By Pythagoras, $H = \sqrt{3^2 + 4^2} = \sqrt{25} = 5$.

$$\sin A = \frac{P}{H} = \frac{3}{5}, \cos A = \frac{B}{H} = \frac{4}{5}, \csc A = \frac{1}{\sin A} = \frac{5}{3}, \sec A = \frac{1}{\cos A} = \frac{5}{4}, \cot A = \frac{1}{\tan A} = \frac{4}{3}.$$

Introduction to Trigonometry — Trigonometric Ratios

- 32. 32.** Prove that $\frac{\tan \theta}{1 - \cot \theta} + \frac{\cot \theta}{1 - \tan \theta} = 1 + \sec \theta \csc \theta$.

Answer:

$$1 + \sec \theta \csc \theta$$

Solution:

1. Convert all terms into terms of $\sin \theta$ and $\cos \theta$.
2. Simplify the fractions by finding a common denominator for the terms.
3. Use the identity $\sin^2 \theta + \cos^2 \theta = 1$.
4. Factor the numerator using the difference of cubes formula $a^3 - b^3 = (a - b)(a^2 + ab + b^2)$.
5. Cancel the common factors and simplify the expression to the final form.

Derivation:

$$LHS = \frac{\frac{\sin \theta}{\cos \theta}}{1 - \frac{\cos \theta}{\sin \theta}} + \frac{\frac{\cos \theta}{\sin \theta}}{1 - \frac{\sin \theta}{\cos \theta}} = \frac{\sin^2 \theta}{\cos \theta (\sin \theta - \cos \theta)} + \frac{\cos^2 \theta}{\sin \theta (\cos \theta - \sin \theta)} = \frac{\sin^3 \theta - \cos^3 \theta}{\sin \theta \cos \theta (\sin \theta - \cos \theta)} = \frac{(\sin \theta - \cos \theta)(\sin^2 \theta + \cos^2 \theta + \sin \theta \cos \theta)}{\sin \theta \cos \theta (\sin \theta - \cos \theta)} = \frac{1 + \sin \theta \cos \theta}{\sin \theta \cos \theta} = \frac{1}{\sin \theta \cos \theta} + 1 = \csc \theta \sec \theta + 1 = RHS$$

Introduction to Trigonometry — Trigonometric Identities

- 33. 33.** If $\sec \theta + \tan \theta = p$, prove that $\sin \theta = \frac{p^2 - 1}{p^2 + 1}$.

Answer:

$$\sin \theta = \frac{p^2 - 1}{p^2 + 1}$$

Solution:

1. Use the identity $\sec^2 \theta - \tan^2 \theta = 1$ to find $\sec \theta - \tan \theta$.
2. Since $(\sec \theta + \tan \theta)(\sec \theta - \tan \theta) = 1$, then $\sec \theta - \tan \theta = 1/p$.
3. Solve the system of two linear equations to find $\sec \theta$ and $\tan \theta$ in terms of p .
4. Calculate $p^2 - 1$ and $p^2 + 1$ using the expressions found.
5. Divide $(p^2 - 1)$ by $(p^2 + 1)$ and simplify to show it equals $\sin \theta$.

Derivation: Given $p = \sec \theta + \tan \theta$. We know $\sec^2 \theta - \tan^2 \theta = 1 \implies (\sec \theta - \tan \theta) = \frac{1}{p}$. Adding equations: $2\sec \theta = p + \frac{1}{p} = \frac{p^2+1}{p} \implies \cos \theta = \frac{2p}{p^2+1}$. Subtracting equations: $2\tan \theta = p - \frac{1}{p} = \frac{p^2-1}{p} \implies \tan \theta = \frac{p^2-1}{2p}$. Since $\sin \theta = \tan \theta \cdot \cos \theta = \frac{p^2-1}{2p} \cdot \frac{2p}{p^2+1} = \frac{p^2-1}{p^2+1}$. *Introduction to Trigonometry — Trigonometric Ratios*

34. 34. If $\sin \theta = \frac{1}{2}$, then the value of $\cos \theta$ is

Answer: (C)

Solution:

1. Use the identity $\sin^2 \theta + \cos^2 \theta = 1$
2. Substitute $\sin \theta = 1/2$
3. Solve for $\cos \theta$

Derivation:

$$\begin{aligned}\cos^2 \theta &= 1 - (1/2)^2 = 1 - 1/4 = 3/4 \\ \implies \cos \theta &= \frac{\sqrt{3}}{2}\end{aligned}$$

Introduction to Trigonometry — Trigonometric Identities

35. 35. The value of $\tan 45^\circ$ is

Answer: (B)

Solution:

1. Recall the trigonometric table for standard angles
2. Identify the value of $\tan 45^\circ$

Derivation:

$$\tan 45^\circ = 1$$

Introduction to Trigonometry — Trigonometric Ratios of Specific Angles

36. 36. If $\cos A = \frac{4}{5}$, then $\tan A$ is

Answer: (A)

Solution:

1. Use $\cos A = \text{base/hypotenuse} = 4/5$
2. Find perpendicular using Pythagoras theorem: $p = \sqrt{5^2 - 4^2} = 3$
3. Calculate $\tan A = p/b = 3/4$

Derivation:

$$\tan A = \frac{\sqrt{1 - \cos^2 A}}{\cos A} = \frac{3/5}{4/5} = \frac{3}{4}$$

37. 37. What is the value of $\sin^2 30^\circ + \cos^2 30^\circ$?

Answer: (C)

Solution:

1. Apply the identity $\sin^2 \theta + \cos^2 \theta = 1$ for any angle θ

Derivation:

$$\sin^2 30^\circ + \cos^2 30^\circ = 1$$

38. 38. If $3 \cot \theta = 2$, then the value of $\tan \theta$ is

Answer: (D)

Solution:

1. Given $3 \cot \theta = 2$, so $\cot \theta = 2/3$
2. Use the reciprocal relation $\tan \theta = 1/\cot \theta$

Derivation:

$$\tan \theta = \frac{1}{2/3} = \frac{3}{2}$$

39. 39. The value of $\frac{\sin 18^\circ}{\cos 72^\circ}$ is

Answer: (B)

Solution:

1. Use complementary angle relation $\cos(90^\circ - \theta) = \sin \theta$
2. Set $\theta = 18^\circ$ so $\cos 72^\circ = \sin 18^\circ$

Derivation:

$$\frac{\sin 18^\circ}{\sin 18^\circ} = 1$$

40. 40. If $\sec \theta = 2$, then θ is

Answer: (C)

Solution:

1. Recall standard values for secant
2. Identify angle where $\cos \theta = 1/2$

Derivation:

$$\begin{aligned}\sec \theta &= 2 \\ \implies \cos \theta &= 1/2 \\ \implies \theta &= 60^\circ\end{aligned}$$

Introduction to Trigonometry — Trigonometric Ratios of Specific Angles